

# CLICK Bench Meter Wiring

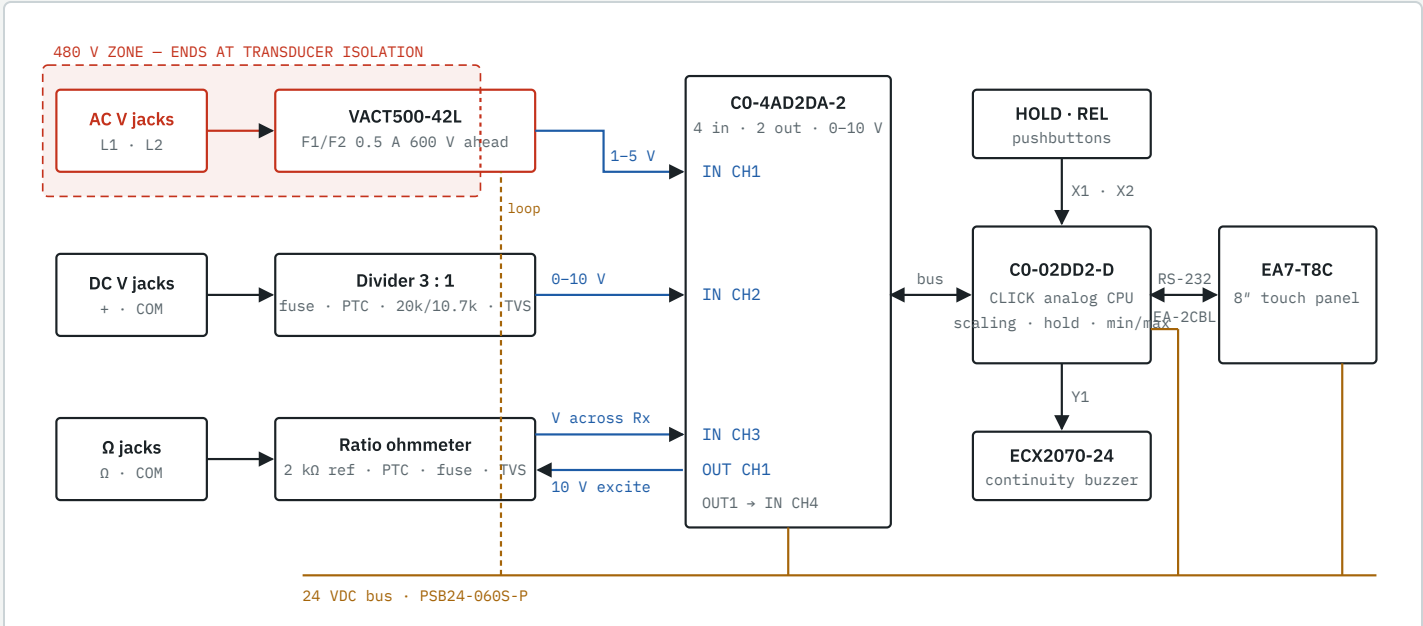
A three-function bench meter (AC volts to 480 V, DC volts to 30 V, ohms and continuity) built from a CLICK C0-02DD2-D PLC, one analog combo module and a C-more EA7-T8C touch panel.

CPU C0-02DD2-D    Analog C0-4AD2DA-2    HMI EA7-T8C    AC transducer VACT500-42L

Supply PSB24-060S-P

— line voltage (up to 480 VAC)    — analog signal (0–10 V)    — 24 VDC power    — return / 0 V

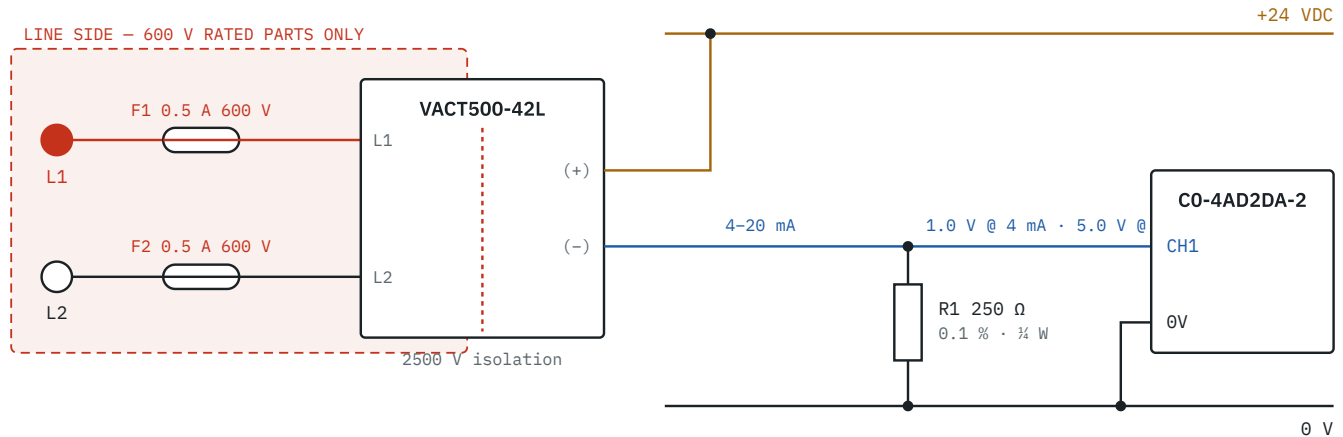
## System overview



**Each function has its own jacks and its own path to its own module input.** Line voltage only reaches the VACT500-42L. Its isolated 4–20 mA output crosses into the low-voltage side. The DC and Ω circuits never share a conductor with the AC jacks, so choosing a mode on the touchscreen can't route 480 V anywhere.

## AC volts – module input CH1

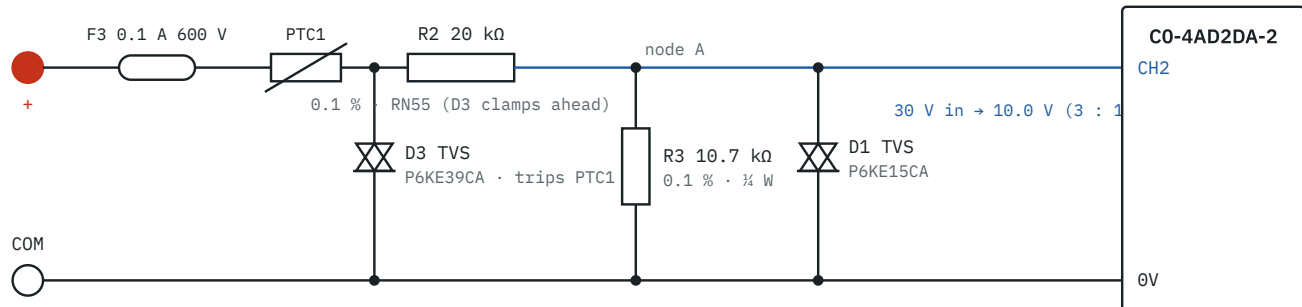
UP TO 480 VAC



The VACT is a 2-wire loop device. +24 V goes into its (+) terminal, and current leaves (-) and flows through R1 to 0 V. CH1 reads the voltage across R1. Use 250 Ω: the datasheet caps the load at 500 Ω, and 250 Ω leaves the transducer plenty of voltage. The output caps at 24 mA (6.0 V), so anything over about 520 V displays as OL. The 0.5 A 600 V fast-acting fuses are the value the VACT datasheet calls for.

## DC volts – module input CH2

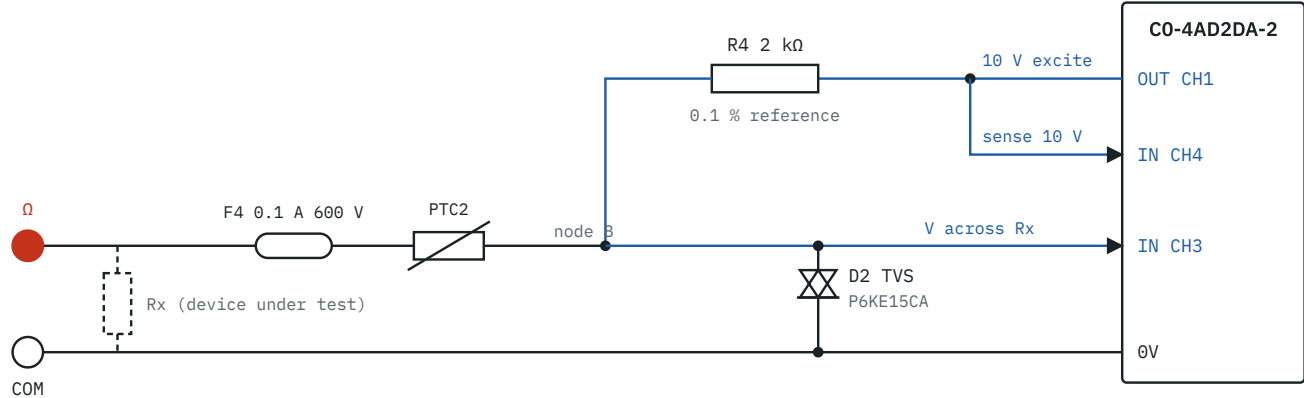
0-30 VDC



**A true 3:1 divider: 30 V in gives 10 V at CH2.** R3 is 10.7 kΩ instead of 10 kΩ because the module's input impedance (about 150 kΩ) sits in parallel with it:  $10.7\text{ k}\Omega \parallel 150\text{ k}\Omega \approx 9.99\text{ k}\Omega$ , which makes the ratio 3.002:1. The spec sheet only guarantees "over 150 kΩ", so check the ratio once against a known meter and trim the  $\times 3.000$  factor in the ladder if it's off. **Fault protection:** if someone puts 480 V on these jacks, D3 clamps right behind PTC1 and pulls amps, so PTC1 trips (or F3 blows) in milliseconds. Without D3, R2 would limit the fault to about 24 mA, too little to trip anything, and R2 would cook at about 11 W. D1 is a second clamp right at the module. This is protection, not a rating: the input is 30 V.

## Ohms & continuity – CH3, CH4, OUT CH1

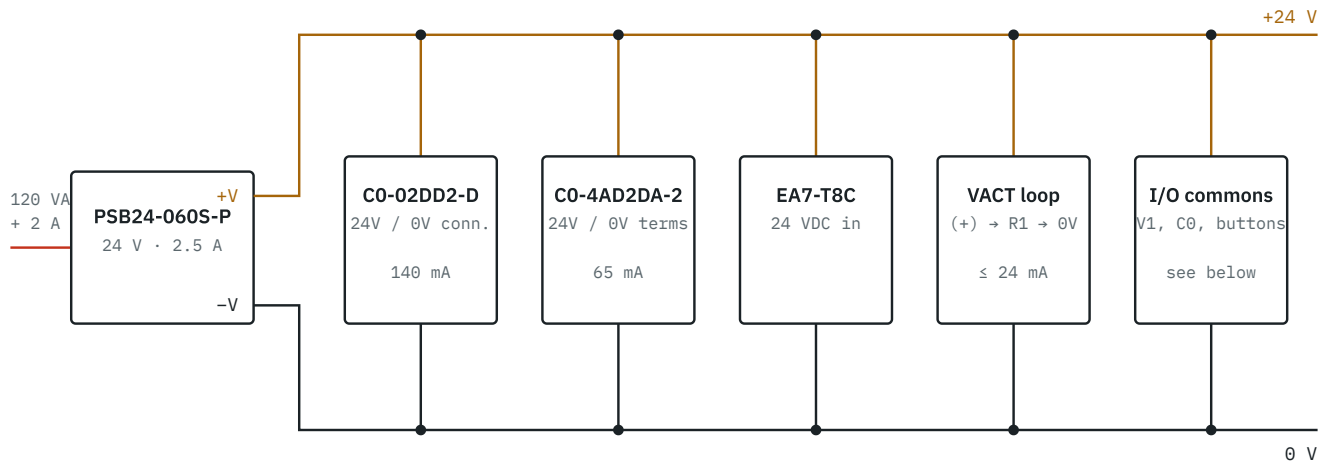
$\approx 1 \Omega - 200 \text{ K}\Omega$



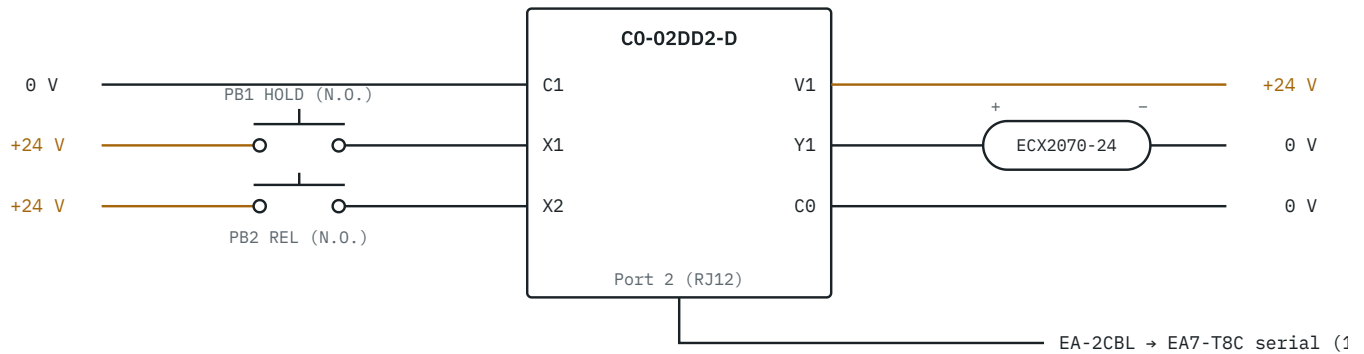
**This is a ratio ohmmeter.** OUT CH1 drives 10 V through R4 into the unknown, and CH4 measures the real excitation. The math uses the ratio of CH3 to CH4, which cancels the output's  $\pm 1\%$  error. Short-circuit current is 5 mA, well under the output's 10 mA limit. PTC2, F4 and the leads add a few ohms, so short the leads and press REL to null them.

## Power, buttons, buzzer, panel link

24 VDC



**Everything runs off one 24 VDC supply.** The total draw is well under 1 A including the panel, so the 2.5 A PSB24 has plenty of headroom. The VACT datasheet calls for a **Class 2** supply. Check the PSB24-060S-P's listing, and if it isn't Class 2, put a 4 A fuse in the VACT's +24 V lead (Class 2 limits at 24 V are about 100 VA / 4 A).



**Inputs X1–X4 share C1.** Tie C1 to 0 V and the buttons switch +24 V in. **Y1–Y4 are sourcing transistor outputs** (0.1 A per point). Feed V1 +24 V, tie C0 to 0 V, and Y1 switches +24 V to the buzzer. **The panel goes on Port 2,** so Port 1 stays free for programming and you never have to unplug the panel to download. In *CLICK Setup* → *Com Port Setup*, set Port 2 to Modbus RTU, node 1, 38400 baud, 8-O-1 to match the panel.

## Point-to-point wire list

| FROM                                           | TO                                                                              | NOTES                                  |
|------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------|
| <b>POWER</b>                                   |                                                                                 |                                        |
| PSB24 +V                                       | CPU 24V · module 24V · T8C 24V · VACT (+) · CPU V1 · PB1/PB2 one side           | +24 V rail, 18 AWG                     |
| PSB24 –V                                       | CPU 0V · module 0V (power) · T8C 0V · R1 low end · CPU C1 · CPU C0 · buzzer (–) | 0 V rail                               |
| 120 VAC cord                                   | PSB24 L / N / $\perp$                                                           | 2 A fuse in line, ground the enclosure |
| <b>AC VOLTS – 600 V RATED WIRE &amp; PARTS</b> |                                                                                 |                                        |
| AC jack L1 (red)                               | F1 → VACT L1                                                                    | F1 KLKD.500 (0.5 A 600 V) in EHM1DU    |
| AC jack L2 (black)                             | F2 → VACT L2                                                                    | F2 KLKD.500 (0.5 A 600 V) in EHM1DU    |
| VACT (–)                                       | module IN CH1 + R1 high end                                                     | R1 250 $\Omega$ 0.1 %                  |
| R1 low end                                     | module IN 0V (next to CH1)                                                      |                                        |
| <b>DC VOLTS</b>                                |                                                                                 |                                        |
| DC jack + (red)                                | F3 → PTC1 → D3 to 0 V → R2 20 k $\Omega$ → node A                               | D3 sits between PTC1 and R2            |

| FROM                         | TO                                                    | NOTES                                       |
|------------------------------|-------------------------------------------------------|---------------------------------------------|
| node A                       | module IN CH2 · R3 10.7 k $\Omega$ to 0 V · D1 to 0 V |                                             |
| DC jack COM (black)          | module IN 0V (next to CH2)                            |                                             |
| <b>OHMS &amp; CONTINUITY</b> |                                                       |                                             |
| module OUT CH1               | R4 2 k $\Omega$ → node B · jumper to IN CH4           | CH4 senses excitation                       |
| node B                       | module IN CH3 · D2 to 0 V · PTC2 → F4 → $\Omega$ jack |                                             |
| $\Omega$ jack COM (black)    | module IN 0V (next to CH3)                            |                                             |
| module OUT CH2               | —                                                     | not connected                               |
| <b>CONTROLS &amp; COMMS</b>  |                                                       |                                             |
| +24 V → PB1 HOLD             | CPU X1                                                | N.O. pushbutton                             |
| +24 V → PB2 REL              | CPU X2                                                | N.O. pushbutton                             |
| CPU Y1                       | ECX2070-24 (+)                                        | buzzer (–) to 0 V                           |
| CPU Port 2                   | T8C serial port                                       | EA-2CBL · Port 1 stays free for programming |

## Small parts and the scaling math

### Digi-Key parts

|         |                                                                       |
|---------|-----------------------------------------------------------------------|
| R1      | RN55E2500BB14 · 250 $\Omega$ 0.1 %                                    |
| R2      | RN55E2002BB14 · 20 k $\Omega$ 0.1 %                                   |
| R3      | RN55E1072BB14 · 10.7 k $\Omega$ 0.1 %                                 |
| R4      | RN55E2001BB14 · 2 k $\Omega$ 0.1 %                                    |
| D1, D2  | P6KE15CA Littelfuse · TVS, 12.8 V standoff                            |
| D3      | P6KE39CA Littelfuse · TVS, 33 V standoff                              |
| PTC1, 2 | TR600-150 Littelfuse · 150 mA hold, 250 V operating / 600 V interrupt |
| F3, F4  | KLKD.100 Littelfuse · 0.1 A 600 V midget, in EHM1DU holders           |
| F1, F2  | KLKD.500 Littelfuse · 0.5 A 600 V midget                              |
| Jacks   | 72913-2 red ×3, 72913-0 black ×3 · Pomona, CAT III 1000 V             |

### Ladder scaling (volts at each input)

```
ACV = (V_CH1 - 1.000) × 125      → 0-  
      OL if ACV > 520
```

```
DCV = V_CH2 × 3.000              (trim  
      OL if V_CH2 > 9.98
```

```
 $\Omega$  = 2000 × V_CH3 / (V_CH4 - V_CH3)  
      OL if V_CH3 > 0.99 × V_CH4
```

```
BEEP (Y1) =  $\Omega$  < 30
```

Inputs are 13-bit (1.22 mV/count). Expect about 0.15 V/count and  $\pm 1\%$  on AC (the VACT's rating), about 4 mV/count on DC, and the best ohms accuracy from 10  $\Omega$  to 50 k $\Omega$ . Drive OUT CH1 at full scale (10 V).

### Before you power it up

- Only the AC jacks, F1/F2 and the VACT input are rated for 480 V. Use 600 V wire there and keep it physically apart from the low-voltage wiring.
- The DC and  $\Omega$  COM jacks are tied to the 24 V 0 V rail. If you earth that rail, those black jacks are at earth, the same as a bench scope's ground clip.
- Test the AC channel first on 120 V against a known-good meter before you go anywhere near 480 V.
- Keep a CAT III/IV handheld for live troubleshooting. This meter is for the bench.